

Proxy Hardening — Test-Type Coverage Status

Revision: 7 **Last modified:** 2026-07-02T16:33:00Z **Status:** In progress — **Rev 7 (2026-07-02): the P10 real-VPN egress-half operator gate is RESOLVED — live Mullvad WireGuard egress is now PROVEN** (persistent one-device Mullvad identity registered + stored §11.4.10; gluetun in rootless podman brought up a kernelspace WireGuard tunnel via relay cz-prg-wg-101 and am.i.mullvad.net/json returned mullvad_exit_ip=true, exit IP 146.70.129.117, Prague CZ; evidence qa-results/verification/mullvad_egress_20260702T161312Z/PROOF.txt). 10 test types PASS with captured live evidence (unit [control-plane, all pkgs pass], stress+chaos, DDoS, concurrency/atomicity, memory-soak, **P10 VPN fail-closed full-automation GREEN, race/deadlock [0 DATA RACE], benchmarking/performance [data-plane p50=86ms + control-plane Go micro-benchmarks], Challenges [2 PASS / 1 honest SKIP], security ACL live-deny GREEN [authoritative Squid access.log TCP_DENIED/403 + HIER_NONE — deny enforced + no upstream leak; S3 Via + S4 SOCKS-SSRF also PASS, wired into the standing suite §11.4.135]**), 1 honest SKIP (HelixQA — bank blocked on 6 un-vendored siblings), integration + e2e PENDING (broader tests/run-tests.sh suite). The control-plane store-layer audit-atomicity TOCTOU was investigated this round and found already CLOSED (store.WithTx), verified RED→GREEN. No PASS is claimed without a real qa-results/ evidence path (§11.4.6). **Authority:** Inherits constitution/Constitution.md per §11.4.35. §11.4.45 integration-status doc reconciling the §11.4.169 test-type-coverage matrix for the proxy hardening workstream. **Companion:** summary [Status_Summary.md](#)

Operator-blocked / pending items (read first — §11.4.45 O(1) surface)

Item	Why blocked / pending	Unblock condition
<i>(none — the P10 real-VPN egress gate is RESOLVED, see below)</i>	—	—

Recently RESOLVED (§11.4.7 demotion-with-evidence): P10 real-VPN egress-half — PROVEN 2026-07-02. The operator gluetun-WireGuard-credential gate that previously blocked this row is cleared: a **persistent one-device Mullvad WireGuard identity** was registered against the operator's Mullvad account via the current app API (POST /auth/v1/token → POST /accounts/v1/devices; the legacy /wg endpoint is deprecated), device name **"groovy rabbit"**, id df4184d4-b6e9-4c2e-b7f7-16e15d4e55a2, and the credentials (VPN_SERVICE_PROVIDER=mullvad, WireGuard key + assigned address) are stored in the **gitignored .env** — never in git (§11.4.10; a §11.4.10.A pre-store leak audit ran CLEAN — 0 tree/history leaks — before storing). **Live egress PROVEN:** gluetun in rootless podman (§11.4.161) brought up a kernelspace WireGuard tunnel via relay cz-prg-wg-101 and https://am.i.mullvad.net/json returned mullvad_exit_ip=true (exit IP 146.70.129.117, Prague, Czech Republic). "One same device" is guaranteed by reusing the persistent private key. Host routing was never disturbed (§11.4.174) and the verification container was torn down (§11.4.14). Evidence: qa-results/verification/mullvad_egress_20260702T161312Z/PROOF.txt. **Honest boundary (§11.4.6):** this proves the Mullvad egress identity + a live tunnel via gluetun (the §11.4.21 credentials gate); wiring the full dynamic proxy-stack data-plane THROUGH gluetun (proxy → tunnel → exit) end-to-end + the up→drop tcpdump no-leak/killswitch half remain the next data-plane step (still tracked in CONTINUATION §3 OWED-TO-P10).

§11.4.169 test-type-coverage matrix (captured-evidence-driven — §11.4.5/§11.4.69)

Base proxy stack UP on :53128 at capture time. Every PASS below cites a real, non-empty qa-results/ artefact captured live this session.

Test type	Status	Evidence path	Notes
Unit	PASS	qa-results/unit/control-plane_unit_20260701T131306Z.log	go test plane: a 100%, 81.2%, 64.5%, only).
Integration	PASS	qa-results/integration/control-plane_verbose_20260702T125403Z/integration_verbose.log	go test plane a honest TestInt interna UsersAn

Test type	Status	Evidence path	Notes
E2E	SKIP (§11.4.3)	—	NoSQLIn (Status PubSub) (8.4s). Healthd Compile cancel. *RealPo contain only, N RealHel honest marked — the c No ded (only s constit Full-au (PASS - Integr API→Po compos operato networ PASS. A NEW w tests/d booted ERR_TUN marker PID un polarity Egress (§11.4. mullvad 146.70. wg-101) full pro step.
Full-automation (P10 VPN fail-closed)	PASS	qa-results/dynamic/vpn_failclosed/20260701T130115Z/verdict.txt (+ req_1..3.body branded pages)	challen proxy: direct), cache \$ Age cap §11.4.3
Challenges	PASS (2/3, 1 honest SKIP)	qa-results/challenges/20260701T131440Z/summary.txt	Honest — subm not ver llm_pro
HelixQA	SKIP (blocked)	qa-results/helixqa/20260701T090532Z/(honest SKIP path)	

Test type	Status	Evidence path	Notes
DDoS / load-flood	PASS	qa-results/ddos/ proxy_flood_20260701T122320Z/flood.evidence (+ census.txt)	Proxy h cases F 6 sibil 300/30 timeout listenin contro row ab server loadflo SSE ov interna 40k cor NumGo read fo review plane_< S1 ACI access example deny CO HIER_NO (no lea supple ambigu allowec HIER_DI Via/ver wired i test_se 6e3e031 group- critical S4 "GR S4(SO0 broken S1&&S one; ru regress acl_gro securit RED_M bearing Stress: concur proxy→ recover contro \$11.4.0
Security	PASS	qa-results/security/ proxy_acl_20260701T152442Z/ s1_acl_deny.evidence	
Stress + Chaos	PASS	qa-results/stress/ proxy_forward_20260701T120843Z/ stress.evidence · qa-results/chaos/ proxy_restart_20260701T120941Z/ recovery_trace.log	

Test type	Status	Evidence path	Notes
Concurrency / atomicity	PASS	qa-results/concurrency/ proxy_concurrency_20260701T122740Z/ concurrency.evidence (+ clients.tsv) · qa- results/unit/control- plane_toctou_atomicity_20260701T131522Z.log	internal rollback recover real-PC aclhelp sustain decisio bearing Decide review plane_< Data-p socks5 OVERALL layer T fails) is mutatio BeginT TestAto polarity go test incl. cn compil {aclhel EXIT= spawn (intern tunnelb layer a round i atomic RSS-bo 000 B, contro — the api/soa proces N-invar 1.5) + audit-a persist gorout review control Data-p live vs p95=0.0 OVERALL now clo
Race / deadlock	PASS	qa-results/race/control- plane_race_20260701T125739Z.log	
Memory	PASS	qa-results/memory/ proxy_soak_20260701T122643Z/soak.evidence (+ rss_timeseries.tsv)	
Benchmarking / performance	PASS	qa-results/benchmark/ proxy_forward_20260701T130414Z/latency.txt · qa-results/benchmark/control- plane_gobench_20260701T131745Z.log	

Test type	Status	Evidence path	Notes
			(0 alloc Compile µs/op.

Honest boundary (§11.4.6)

This matrix reconciles the **hardening** test-type picture proven in this session. Ten types (unit, stress+chaos, DDoS, concurrency/atomicity, memory, P10 fail-closed full-automation, race/deadlock, benchmarking, Challenges, security ACL live-deny) are PASS with captured live qa-results/ evidence. The security-ACL row moved SKIP→PASS this round: the 000 client-code ambiguity that previously forced the SKIP was replaced by reading Squid's own access.log (TCP_DENIED/403 + HIER_NONE), the authoritative sink-side oracle (§11.4.69), with a RED teeth-check (allowed :443 → TCP_TUNNEL/HIER_DIRECT → no false deny-PASS, §11.4.115) and the guard wired into the standing suite (§11.4.135). One SKIP is honest (§11.4.3): HelixQA (bank blocked on 6 un-vendored own-org siblings — the runner + proxy.yaml bank exist; unblock = vendor them or pass HELIXQA_BIN). Integration + e2e remain PENDING — not claims the capability is broken; covered by the broader tests/run-tests.sh suite. The two follow-ups flagged earlier this round are RESOLVED: the store-layer audit-atomicity TOCTOU was found already CLOSED (store.WithTx, verified RED→GREEN), and the control-plane Go-benchmark gap is closed (3 new *_bench_test.go). **Rev 7 (§11.4.7):** the P10 real-VPN egress-half is no longer an operator-gated SKIP — a persistent one-device Mullvad WireGuard identity is registered + stored (§11.4.10, .env only) and live Mullvad egress is PROVEN via gluetun (mullvad_exit_ip=true, evidence qa-results/verification/mullvad_egress_20260702T161312Z/PROOF.txt); the remaining honest boundary is the full proxy-data-plane-through-gluetun end-to-end path. None of this substitutes for the §11.4.40 full-suite retest before a release tag.